

AMERICAN COLLEGE OF SCIENCE AND TECHNOLOGY

ACOSAT -SL



COURSE PAMPHLET

INTRODUCTION TO COMPUTER BASICS WINDOWS

LECTURER

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AN INTRODUCTION TO COMPUTER BASICS

MS WINDOW(S)

Introduction

Computers have become one of the most essential parts of human life. At present, computers can be easily seen in almost every sector or field even where it is most unexpected. There can be several different reasons why computers are actually required. We can summarize the reasons for the requirement of computers in three words: **efficiency, accuracy, and reliability.**

Nowadays, computers are making jobs easier for people. Computers can be used in everything from entertainment to communication to navigation to research. That is why this era is called the era of **IT (Information Technology)**. And now, one cannot imagine a world without computers.

Therefore, it is very necessary to have knowledge of Computer basics. In this two weeks lesson, we are look at the basics of the computer such as **what is a computer, definition of computer, basic functions of a computer, booting and shutdown, classification or types of computer, advantages, and disadvantages of a computer, some uses of computer, Hardware vs Software, components of a computer, the window environment, etc.**

The Term Computer

In 1640, the term '**Computer**' was initially referred to as '**one who calculates**'. Later in 1897, it was called the '**Calculating Machine**'. In 1945, it was indicated as '**programmable digital electronic computer**' which is now called a 'computer'.

The term 'Computer' is derived from the Latin word '**computare**', which is defined as- "to calculate", "to count" or "to sum up", etc. In other words, "**a computer is a device that performs computation**".

Definition of a Computer

“A computer is a programmable electronic machine designed to take input, perform prescribed arithmetic and logical operations at fast speeds, and provide the output of these operations.”

The term '**COMPUTER**' is an acronym for '**Common Operating Machine Purposely Used for Technological and Educational Research**'.

Note- The first mechanical computer was designed in 1837 by **Charles Babbage**. It was called 'Analytical Engine'. It was the first general-purpose computer. **Charles Babbage** is known as the father of the computer.



Figure 1: Personal Computer

Basic Functions of Computer

There are four basic functions of the computer: **Input, Processing, Output, and Storage.**

Input: The data is entered into the computer with the help of input devices. Like other electronic devices, a computer takes data in raw form (binary form). The user can enter

the data in several formats such as the collection of letters, numbers, images, etc. The input devices convert the data in the binary form so that the computer can read the data.

Some of the main input devices of computer systems are:

- Keyboard
- Mouse
- Joystick
- Scanner
- Microphone
- Lightpen

Processing: The processing is the core functionality of the computer system. It is the internal process where the data is processed according to the instructions given to the computer. The data is executed sequentially and sent for further processing.

The processing speed may vary in different computer systems as the speed mainly depends upon factors like which type of Motherboard, **CPU (Central Processing Unit)** or **RAM (Random Access Memory)** you are using.

Output: The output is the information provided by the computer after the entire processing. It is also known as the result that can be stored in the storage devices for further use. The output devices retrieve the processed data from the computer and convert the data into a human-readable form.

The widely used output devices of computer systems are:

- Monitor
- Printer
- Projector
- Speakers

Storage: Storage is a crucial part of the computer system. It is used to store data or instructions before and after processing.

There are two types of storage: **Primary Storage and Secondary Storage**

Primary storage devices store the inputted data and immediate calculation results. The data stored in primary storage is temporary and will be lost if they are disconnected from the power source.

Random Access Memory is an example of primary storage.



Secondary storage devices are used to store the data permanently for future use. The data stored in the secondary storage devices is secure even if there is no power supply.

Hard Disk Drive is the widely used secondary storage.

Bootting /Boot / Start-Up / Power-On

Bootting means to start up your computer that is giving power to the system. When you power on a system, wait till the computer self-start to point. During this time, the computer will under a process known as POST, short for **Power on Self-Test** where the computer will test itself for all the peripherals that are connected to the system **unit**. (Input and output devices)

How to Boot a Computer (Desktop Computer)?

The computer needs to be turned on to work. To turn on the computer system, you must do the following.

- 1) Check the cables connected to the system unit
- 2) Check the main socket connection
- 3) Remove all removable storage devices from the system unit
- 4) Press the power button of the monitor
- 5) Press the power button on the system unit

How To Shutdown A Computer

To **shut down** or power off a **computer** is to remove power from a **computer's** main components in a controlled way. ... Shutting down a computer closes all the programs that are open and exits the operating system safely so hardware isn't damaged, and your

operating system isn't corrupted. The following article describes how to shut down a computer, with instructions for different operating systems.

All versions of Microsoft Windows can be shut down through the Start Menu.

- 1) Press the Windows key on the keyboard or click the Start button.
- 2) From the Start Menu, click **Shut Down...**
- 3) Wait for few minutes, and Turn off the Monitor by pressing the power button

Advantages and Disadvantages of Computer

Just like a coin, a computer system also has two sides: Advantages and Disadvantages. While there are several disadvantages of a computer system, but, the advantages overcome them. Let's discuss both advantages and disadvantages of the computer system:

Advantages of Computer

The main advantages of the system are listed below:

- Amazing Speed
- Accuracy
- Huge Storage
- Multitasking Support
- Data Security
- Automation
- Reduced Cost

Disadvantages of Computer

The main disadvantages of the computer system are listed below:

- Unemployment
- Health Issues
- Cyber Crimes
- Virus and Hacking Attacks
- Improper Use

- Spread of False or Inappropriate Content
- Negative Impact on the Environment

Some Uses of Computer

As we have discussed above, the computer is the need for the digital era. It is helping people to complete their tasks in hours that were before taking several days or months.

Here, we have explained the most important uses of the computer system:

- Computer can store important data.
- Computer can communicate with people around the world.
- Computer can be used for the educational system.
- Computer can be used for office works like data entry, support, development, etc.
- Computer can be used in banking, finance, and marketing.
- Computer can be used in government sectors.
- Computer can be used in sports.
- Computer can be used for Press and publishing.
- Computer can be Used for entertainment.

Classification of Computer

According to usage and functionalities, computers can be classified as:

Analog Computer – The computers used to process analog data (continuously varying data) are called ‘Analog Computers’. Analog computers are the most complicated machines for computation and process control. Examples of continuous data are pressure, temperature, voltage, weight, and speed, etc.



Speedometer

Digital Computer – Digital computers are personal computers. These are the widely used computers. These are commonly used for processing the data with a number using digits by utilizing the binary number system. These computers are intended to perform arithmetic and logical operations at a very high rate.



Hybrid Computer – Hybrid computers are the combination of analog computers and digital computers. These are as fast as analog computers and include memory and precision as digital computers.

Types of Computer

Supercomputer – Supercomputers are large and require huge space for the installation. They are the fastest and most expensive computers compared to others. They are used for performing huge complex calculations.



Mainframe Computer – Mainframe Computers are smaller than supercomputers, still, they comparatively huge. These are not as fast as supercomputers. These are expensive as well as take huge space for the entire setup. Mainframe computers can store huge amounts of data and they are capable to handle large calculations. These computers are usually found in banks and educational sectors.



Mini Computer – A **minicomputer** is also known as mini. It is a class of small computers that was introduced into the world in the mid-1960s.

A minicomputer is a computer which has all the features of a large size computer, but its size is smaller than those. A **minicomputer** lies

between the mainframe and the

microcomputer because its size is smaller than the former one and larger than the latter one



Microcomputer – Microcomputers are inexpensive and support multi-user platform. These types of computers are mostly used by small organizations. Microcomputers are slower compared to supercomputers and mainframe computers. Microcomputers are called Personal Computer (PC).



Computer Hardware, Software, and User

Computer Hardware are the physical components or parts that jointly form a computer system. There are different types of hardware. Depending on the structure of the computer system, hardware can be installed inside or outside of the computer physical body.

The basic components of the computer hardware are listed below:

- Monitor
- Keyboard
- Mouse
- CPU (Central Processing Unit)
- Motherboard
- RAM (Random Access Memory)
- Hard Disk Drive
- Printer, etc

Software is a set of instructions known as “**Programs**” that controls the computer hardware, it is been coded either by the programmers for the computer hardware to follow. It deals with the operational aspect of the computer studies

Software can be of two types.

- System Software
- Application Software.

Systems Software are programs that manage the resources of the computer system and simplify applications programming. They include software such as the operating system, database management systems, networking software, translators, and software utilities.

Application Software, or app for short, is software that performs specific tasks for an end-user. For example, Microsoft Word or Excel are application software, as are common web browsers such as Firefox or Google Chrome.

Software Programs

Many software programs can be installed into the computer system depending on the type of operation you want to perform or the business you want to do. This software includes:

Operating System: Like Disk Operating System (DOS), Linux, Unix, Windows (Windows XP, Windows Vista, Windows 7, 8, 8.1, 10, 10 Pro, and Windows 11). etc.

Office Programs: office 360, office 2007, office 2010, office 2013, office 2016, and office 2019. Product of office programs are suite comprise of up to five (5) programs which are: **MS Word, MS Excel, MS Access, MS PowerPoint, MS Publisher**, etc.

Game Software: FIFA, GTA-5, Mortal Kombat, Spider Man, Injustice, Call of Duty, etc.

Anti Virus Software: McAfee , Norton , Panda, TotalAV, Bitdefender, Kaspersky , AVG. AVG is one of the most popular antivirus programs that can be obtained for free, and it's easy to download directly from the internet.

User/Live Ware

This refers to the user of the computer machine...the following can be referred to as users:

A Computer Professional - is a person who has a formal education in the technical aspect of using a computer. E.g., computer programmer designs write and implement the programs that process the data on the computer.

A computer user (or end-user) is a person like you – someone without much technical knowledge of computers but who uses the computer to perform professional or personal tasks, enhance learning, or having fun.

Main Component / Hardware of a Computer System

The System Unit –

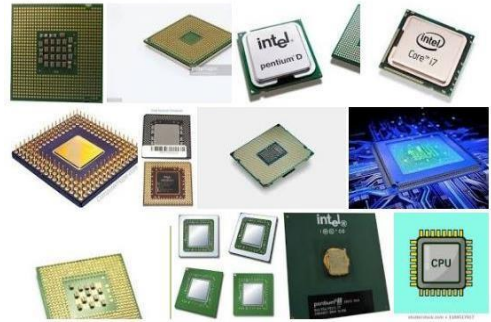
The System Unit is NOT the CPU as some people call it. It is the block that houses the basic hardware components in the computer system like the; Motherboard, Processor (CPU), Power Supply, Memory, Disk Drives, etc.



The Central Processing Unit (CPU) / PROCESSOR

The CPU inside the system unit is known as the **brain** of the computer and can be named as **Central Processing Unit** or simply the

PROCESSOR. This is the unit that is responsible for all the processing, storing, and retrieving of all information as the computer is defined. The CPU is divided into three main parts; the **Control Unit**, the **A.L.U** (Arithmetic and Logical Unit) and the **Primary Storage**.



The Monitor/VDU

A monitor or screen is the most common output device associated with a computer, which displays video, images, and texts. A monitor resembles a television and normally displays information at a much high resolution. A monitor generally can connect to a VGA and HDMI port. we called it to monitor because the users sit by it and monitor his or her work. As an output device, it displays softcopy information on the screen as you type or activate any program or instruction. The monitor can be of two types.

- **Cathode-Ray Tube Monitors (CRT),**
- **Flat-Screen Monitors (Liquid Crystal Display LCD)**



Cathode-Ray Tube Monitors (CRT), seeing in most desktop computers. They are usually large in size, heavier in weight than the liquid crystal display unit, and consume large amounts of electricity.



Flat-Screen Monitors are found on most laptop computers, personal digital assistants (PDAs), multimedia devices, digital cameras, and several desktop computers. LCD monitors are thinner, light-weighted, energy effective, and smaller in size.

They have virtually replaced than the CRT monitors. They are mostly found on laptops, calculators digital watches, digital cameras, some desktop computers, and other similar devices.

The Computer Keyboard



Keyboard is the most common and very popular input device which helps to input data to the computer. The layout of the keyboard is like that of traditional typewriter, although there are some additional keys provided for performing additional functions.

Keyboards are of two sizes 84 keys or 101/102 keys, but now keyboards with 104 keys or 108 keys are also available for Windows and Internet:

The keys on the keyboard are as follows;

Typing Keys : These keys include the letter keys (A-Z) and digit keys (09) which generally give the same layout as that of typewriters.

Numeric Keypad : It is used to enter the numeric data or cursor movement. Generally, it consists of a set of 17 keys that are laid out in the same configuration used by most adding machines and calculators.

Function Keys: The twelve function keys are present on the keyboard which are arranged in a row at the top of the keyboard. Each function key has a unique meaning and is used for some specific purpose.

Control keys: These keys provide cursor and screen control. It includes four directional arrow keys. Control keys also include Home, End, Insert, Delete, Page Up, Page Down, Control(Ctrl), Alternate(Alt), Escape(Esc).

Special Purpose Keys : Keyboard also contains some special purpose keys such as Enter, Shift, Caps Lock, Num Lock, Space bar, Tab, and Print Screen.

The Computer Mouse



The computer mouse is designed to help the computer user to point to objects on the computer screen, select them and move them about.

Advantages

- Easy to use
- Less expensive
- Enhance the cursor movement faster than the arrow keys of the keyboard.

The mouse can be used to perform five different operations, these include: **Pointing, Single Clicking, Double Clicking, Right-Clicking, and Drag & Drop**

Pointing is usually not considered as a mouse skill because it is regarded as an easy thing to do, yet for many new computer users, this can be a very skill to master.

Pointing simply refers to the movement of the mouse pointer to a location on the computer screen.

Effective pointing requires two things. First, you must hold the computer mouse properly. Make sure that the mouse fits inside the curve of your palm.

The second thing associated with pointing is to make sure you do not run out of mouse space. Skilled mouse users can move the mouse pointer without having to move the mouse across a large area on the desk.

Single clicking is also known as clicking. Clicking or single-clicking refers to the process of pressing and releasing the mouse button. Clicking is usually associated with the left mouse button. This is known as left-clicking.

Double-clicking refers to the process where you click the left mouse button twice in quick succession in a particular spot.

Drag and drop references to the process of pointing to an object, pressing the left mouse button, and holding it down whilst moving the mouse and the selected object to a new location, then release the button. Drag and drop are used to move files and folders as well as icons from one location to another.

Printers

These are the output devices that print documents as they are created or activated on the screen.

Printer: A printer is an output device mostly found in offices and homes. It produces a permanent copy known as hardcopy from a softcopy of any work stored in a computer or computing device, meaning outputting images and texts onto a paper. There are two different types of printers:

Impact Printers and non- Impact Printers. Several modern printers connect to a computer using a USB cable or via Wi-Fi.



Printer.

There are two different types of printers:

- **Impact Printers**
- **Non- Impact Printers**



Impact Printers - Impact printer uses hammers, pins, or plastic heads to strike against a ribbon and paper to print the text. This mechanism is known as the electromechanical mechanism. E.g., DOT- Matrix, Line printer.



Non-Impact Printers: This type of printer prints without hitting a ribbon onto a page. It produces better quality at high speed. An example: includes Ink Jet printers and LaserJet printers.

The Speaker

Speakers are one of the most common output devices used with **computer** systems. Speakers are essential if you want a louder sound, surround sound, fuller bass, or just a better quality of audio. Regardless of their design, the purpose of **speakers** is to produce audio output that can be heard by the listener. Some computer speakers are wireless nowadays, connecting to the computer via Bluetooth.



The speaker is an output device that is used to produce softcopy output in the form of voice.

Headphones



Headphone

Headphones are called earphones and they permit you alone to listen to audio sound without disturbing other people. They connect via the computer line out, or to the speakers. It was initially used by the US Navy in 1910 and today

almost used by all works of life. They come in all sorts of shapes and sizes, from basic earbuds to the more traditional style with padding around the earpieces and a connecting band that fits over the user's head. **projector**



A **projector** is an output device that transmits computergenerated images and texts onto a wall, screen, or another surface. Projectors are typically used for presentations, watching movies, or as a teaching aid, as they enable an entire roomful of people to see images generated by a single computer. Modern projectors usually connect to the computer via an HDMI cable or

VGA. It can also produce either a still (slides) or moving images (videos).

Microphone



Microphone is an input device to input sound that is then stored in a digital form.

The microphone is used for various applications such as adding sound to a multimedia presentation or for mixing music.

INTRODUCTION MS WINDOW(S) ENVIRONMENT

What do you understand by Microsoft Windows?

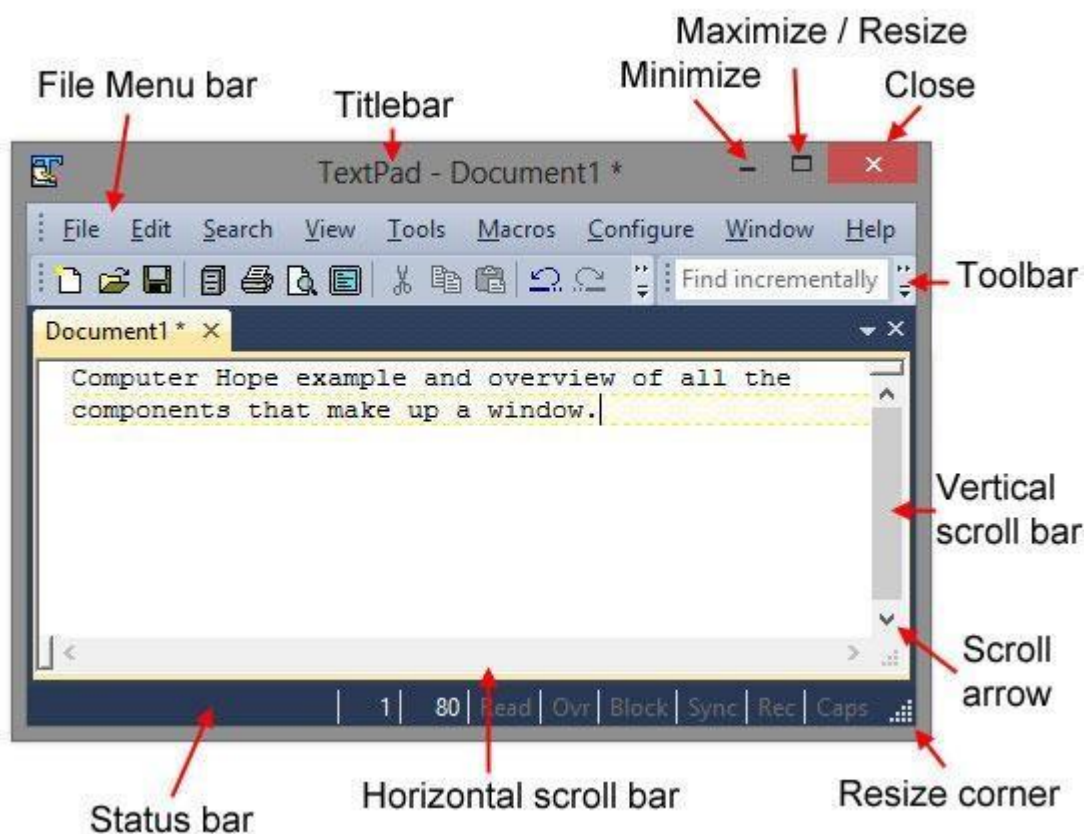
Microsoft Windows, also called Windows and Windows OS, is a computer operating system (OS) developed by Microsoft Corporation to run personal computers (PCs). Featuring the first graphical user interface (GUI) for IBM-compatible PCs, the Windows OS soon dominated the PC market.

Microsoft created the graphical operating system known as Windows. It offers a way to access the internet, lets users view and store files, run software, play games, and watch videos. It was made available for both personal and business use.

A Window Environment

A window environment is a rectangular section of the computer's display in a GUI(Graphical User Interface) that shows the program currently being used. For example, the browser window you're using to view this web page is a **window**. Windows allow a user to work with multiple programs or view multiple programs at once. Almost all windows allow you to minimize and maximize them, allowing you to hide and view a program temporarily. Below is what a window may look like in Microsoft Windows and each of its major features known as the **window elements**.

Example of a Microsoft Windows window



Note: If a window is not fullscreen (maximized), it's considered to be in window mode. In this mode, the window can be moved around the screen.

Common Windows Items

The following are just some of the terms used to describe the Windows desktop you are likely to run into. Most are indicated in the diagram below the list:

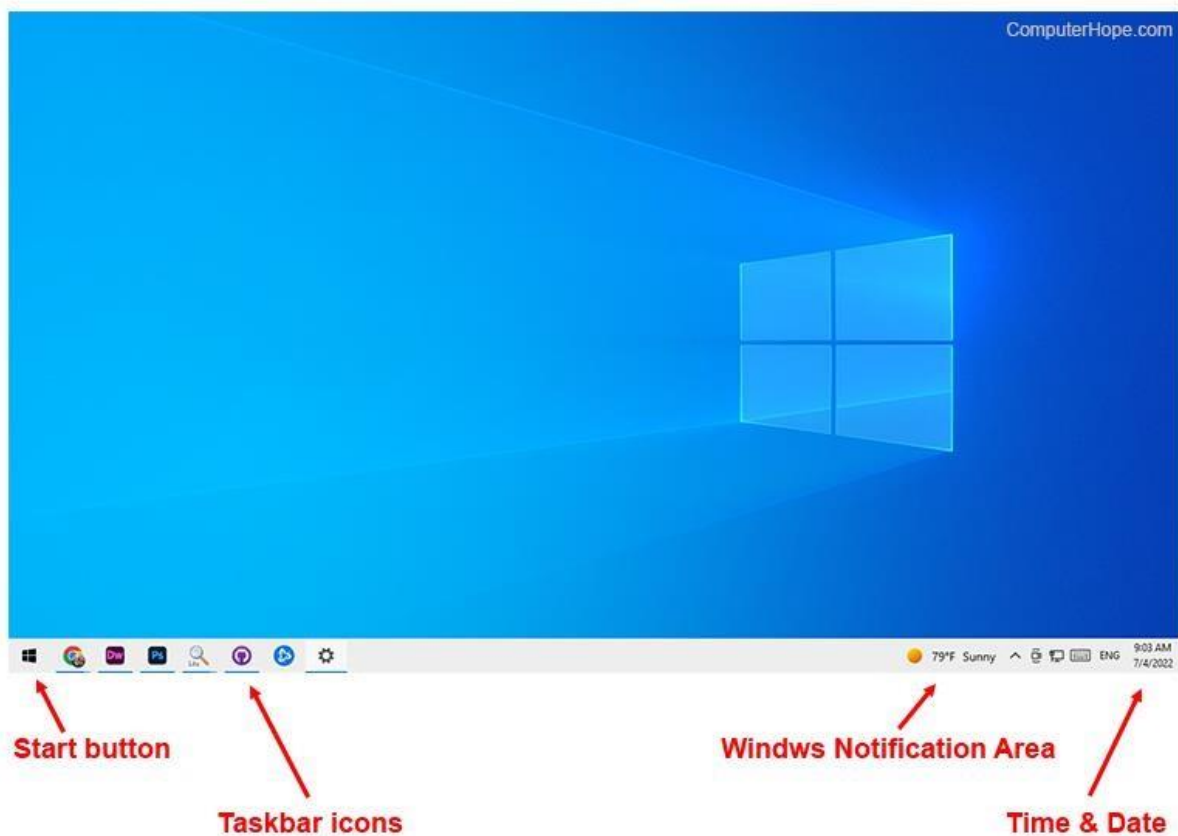
- 1) **Desktop** refers to the background of your screen on which the various programs run. Think of your computer screen as your electronic desktop.
- 2) **Icons** are those small pictures on the desktop and inside folders that represent various programs, specialized folders, etc.
- 3) **Folders** are containers that can contain icons, programs, data or other folders (subfolders). The default folder icon looks like a Manila file folder but the look varies with different Windows versions. System folders generally look like icons and any folder can be modified to change its appearance.
- 4) **Title bar** refers to the bar at the top of an open window. The folder's title will tell you what the folder is used for. It contains the minimize, maximize and close buttons. You can use the title bar to move a window around.
- 5) **Cursor** is the graphic pointer which indicates where the mouse is and what sort of action it is performing. The cursor will change from the default arrow to various shapes according to the purpose it is serving at the time. For instance, it may form an I-beam shape when you are selecting text in a document or a double-arrow when you are resizing a window.
- 6) **Task bar** refers to the bar usually at the bottom of your Windows screen (it is movable) containing the Start Button on the left and the clock and grouped icons on the right.
- 7) A **Scroll Bar** appears when there is more information in the window than can be displayed. This is usually a vertical scroll bar, but a horizontal scroll bar may appear if the width of the window is too narrow. Windows 10 often hides scroll bars until you hover over them.
- 8) The **Address Bar** tells you your location and provides the tools to navigate around your computer. These settings vary by Windows version and can be modified.

While this list is based upon Windows, other operating systems like Mac and Linux use a lot of the same terminology.

Desktop

When referring to an operating system or GUI (graphical user interface), the **desktop** is the primary place to display and organize icons on a screen. The Microsoft Windows desktop was first introduced with Microsoft Windows 95 and has been included with every version of Windows since then. The image below is an example of the Microsoft Windows 10 desktop

Windows 10 Desktop



In this example, there are no icons on the desktop, and the wallpaper is a blue screen with the Windows 10 logo. Also, the taskbar is at the bottom of the desktop and contains the Start, taskbar icons, Windows Notification Area, and the time and date.

What icons and items are found on the Windows desktop?

Common icons on the desktop include those for My Computer, Recycle Bin, your Internet browser (e.g., Internet Explorer), and My Documents. On the Windows desktop, you can find the Windows Start menu through the Start on the taskbar, and the Windows Notification Area.

If some or all of these icons are missing on the desktop, you can change which icons are displayed. Click the link below and follow the steps to show or hide these icons.

COMPUTER LAB RULES

- 1) Do not bring food and drinks to the computer lab.
- 2) Be Safe, Respectful and Responsible
- 3) No downloading games, screen savers, changing the system settings & background.
- 4) Keep your hands to yourself. Don't touch your neighbor's keyboard or mouse unless you're invited.
- 5) Keep your hands off the keyboard and mouse until you and the computer are ready.
- 6) Ask permission before printing
- 7) Work quietly and stay on task. Make every effort to do your best.
- 8) Ask for assistance if you're unsure what to do.
- 9) Never add any software or hardware component to the computer system.
- 10) Do not change any settings on the computer.
- 11) Avoid talking unnecessarily in the lab.
- 12) Make sure to report any software or hardware related problem to your teacher rather than trying to solve it yourself.
- 13) Do not move about in the lab.
- 14) To help your neighbor, talk at a low tone.
- 15) Computers should be turned on/off properly. Use the power buttons to turn on the monitor and system unit respectively but "DO NOT" use the power button to turn off the system unit.

16) Do not delete anything from the computer.

17) Do not connect or disconnect any cable unless your teacher asks you to do so.

Summary

The computer plays an important role in our day-to-day life and we cannot think of life without it. While there are some disadvantages of computers, however, the advantages of computers overcome those disadvantages. Therefore, it is very important to have at least basic knowledge of computers to complete in this era of technology.

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